Selim Molla

PHD CANDIDATE ▪ COMPUTATIONAL SCIENCE PROGRAM

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RESEARCH EXPERIENCE

University of Texas at El Paso, USA September 2023 - Present

Graduate Research Assistant, Dept. of Industrial, Manufacturing, and Systems Engineering

- Conduct research at the "**Research Institute for Manufacturing and Engineering Systems (RIMES)**" in computational modeling, simulation, optimization, artificial intelligence, and data-driven decision-making.
- Develop simulation-based models using FlexSim, FlexSim HC, and AnyLogic for manufacturing, healthcare, and complex operational systems.
- Apply machine learning, deep learning, and reinforcement learning methods to forecasting, monitoring, optimization, and simulation-based decision-making problems.
- Integrate mathematical optimization with discrete-event simulation to support resource allocation, production planning, and system-performance analysis under uncertainty.
- Conduct simulation experiments, statistical and performance analysis, model validation, and research dissemination through peer-reviewed publications and conference presentations.

TEACHING EXPERIENCE

University of Texas at El Paso, USA September 2026 - Present

Co-Instructor, Industrial Systems Simulation course, IMSE Department

- Co-teach the Industrial Systems Simulation course, including lectures, laboratory exercises, simulation modeling activities, and student support.
- Develop and demonstrate simulation models and guide students in applying discrete-event simulation concepts using FlexSim.

University of Texas at El Paso, USA September 2023 - Present

Graduate Teaching Assistant, Computational Science Program

- Assist with undergraduate teaching through review sessions, grading, examination proctoring, and student support.
- **Courses assisted:** Responsible AI for Decision-Critical Systems, Probability and Statistics, Calculus III, Calculus I, Matrix Algebra, Precalculus, Math for Social Sciences I

Khulna University of Engineering & Technology, Bangladesh September 2021 – December 2022

Teaching Assistant, Department of Industrial Engineering and Management

- Assisted with grading script and proctoring examinations

EDUCATION

Ph.D. in Computational Science *in Progress*
University of Texas at El Paso (UTEP), USA.

M.Sc. in Computational Science *Awarded: 12/2025*
University of Texas at El Paso (UTEP), USA. ***CGPA: 3.94/4.00***

B.Sc. in Industrial & Production Engineering *Awarded: 05/2017*
Khulna University of Engineering and Technology (KUET), Bangladesh. ***GPA: 3.29 /4.00***

RESEARCH INTEREST

- Computational modeling, simulation, and optimization of large and complex systems
- Machine learning, deep learning, and reinforcement learning for manufacturing and healthcare applications
- Artificial intelligence and digital twins for intelligent manufacturing systems and decision support
- Data-driven methods for resource allocation, system optimization, and operational decision-making

RESEARCH PROJECTS

M.Sc. thesis title: *"Process-Centric Simulation to Optimize Complex Production Lines Considering Multi-Class Parts Combination."*

- Developed a hybrid simulation-guided mathematical framework (SiGMa) integrating mathematical programming and discrete-event simulation to optimize workforce allocation in mixed-model assembly lines under real-world variability.
- Formulated a mathematical programming model incorporating system constraints and solved it using Python (PuLP).
- Built and validated a FlexSim-based discrete-event simulation (DES) model to verify optimization results and capture system dynamics.
- Applied the approach to an industrial case study, demonstrating improved workforce utilization and production efficiency.

B.Sc. thesis title: ***"****Line Balancing by Linear Programming.****"***

- Formulated and solved an assembly line balancing problem using linear programming to optimize task allocation across workstations.
- Implemented multiple heuristic methods (Largest Candidate Rule, Kilbridge & Wester, Ranked Positional Weights) and compared their performance with the optimal LP solution.
- Developed Python-based code to automatically generate objective functions and constraints, and solved the model using LINDO.
- Demonstrated improved efficiency of linear programming over heuristic approaches in workload balancing and resource utilization.

PUBLICATIONS

Published

- **Molla, S.**, Rahman, M. F., Tseng, T. L. B., Chiou, R. Y., & Quddus, M. A. (2026). A Risk-Free Approach to Validate Drone Flight Paths Using Computer Simulation and Reinforcement Learning. In *Proceedings of the 2026 ASEE Annual Conference & Exposition*.
- Arvizu Astorga, J. F., **Molla, S.**, Tseng, T. L. B., & Rahman, M. F. (2026). A Noble Digital Twin-Based Inspection Scheme (DTIS) for Real-Time Monitoring in Smart Manufacturing. In *Proceedings of the Manufacturing Science and Engineering Conference (MSEC 2026)*.
- Siddique, M. A., **Molla, S.**, Lopes, A. J., & Rahman, M. F. (2025). Simulation-Based Job Shop Scheduling to Minimize Resource Uncertainties in Manufacturing Environment. In *Proceedings of the 2025 Winter Simulation Conference (WSC)*, 1541–1552.
- Rahman, S. M. A., **Molla, S.**, Sultana, J., Chiou, R. Y., Tseng, T. L. B., & Rahman, M. F. (2025). Development of Digital Laboratory Modules Using Computer Simulation for Enhanced Learning Experience in Manufacturing Education. *Manufacturing Letters*.
- **Molla, S.** (2025). *Process-Centric Simulation To Optimize Complex Production Lines Considering Multi-Class Parts Combination*. Open Access Theses & Dissertations, 4571.
- Hasan, M. R., **Molla, S.**, & Siddique, I. M. Next-Gen Production Excellence: A Deep Simulation Perspective on Process Improvement. *Journal of Mechatronics, Machine Design and Manufacturing, 6*(1), 7–20.
- Noman, A. H. M., Mustaquim, S. M., **Molla, S.**, & Siddique, I. M. (2024). Enhancing Operations Quality Improvement Through Advanced Data Analytics. *Journal of Computer Science Engineering and Software Testing, 10*(1), 1–14.
- Ullah, M. R., **Molla, S.**, Siddique, I. M., Siddique, A. A., & Abedin, M. M. (2023). Optimizing Performance: A Deep Dive Into Overall Equipment Effectiveness (OEE) for Operational Excellence. *Journal of Industrial Mechanics, 8*(3), 26–40.
- Siddique, I. M., Siddique, A. A., Smith, E. D., & **Molla, S.** (2023). Assessing the Sustainability of Bitcoin Mining: Comparative Review of Renewable Energy Sources. *Journal of Alternative and Renewable Energy Sources, 10*(1), 1–12.

Accepted Conference Paper

- **Molla, S.**, Raut, S., Hossain, N. U. I., Sokolov, A., Gorod, A., & Rahman, M. F. (2026). AI-Enabled Forecasting for Wind Energy Management. In *Proceedings of the American Society for Engineering Management (ASEM) 2026 International Annual Conference*.

Accepted Book Chapter

- Rahman, S. M. A., **Molla, S**., & Rahman, M. F. (2026). *Systematic Data-Driven Approaches and Applications to Overcome the Risk of Supply Chain Disruption and Mitigation*. Book chapter accepted for publication by Taylor & Francis.

Under Review

- **Molla, S.**, Rahman, M. F., Rahman, S. M. A., Tseng, T.-L. (Bill), Lopes, A. J., & Hossain, N. U. I. (2026). A Simulation-Guided Optimization Framework for Mixed-Model Assembly Lines under Process Stochasticity. *European Journal of Operational Research*. Under review.
- Rahman, S. M. A., **Molla, S.**, Rahman, M. F., Tseng, T.-L. (Bill), Hossain, N. U. I., & Quddus, M. A. (2026). Intelligent Job-Shop Operations Using Digital Twin-Integrated Fallback Routing and Dynamic Dispatching. *Expert Systems with Applications*. Under review.
- **Molla, S.**, Rahman, M. F., & Hossain, N. U. I. (2026). Predicting Blockchain-Enabled Information Integration in Healthcare Supply Chain. *Smart Health*. Under review.

RESEARCH PRESENTATIONS

- **Technical Presentation**, ASME Manufacturing Science and Engineering Conference (MSEC 2026), The Pennsylvania State University, State College, PA, 2026.
- Poster Presentation, IMSE Day, 2026, UTEP
- Poster Presentation, IMSE Day, 2025, UTEP
- Poster Presentation, IMSE Day, 2024, UTEP

PEER REVIEW EXPERIENCE

- **Reviewer for *International Journal of Research in Industrial Engineering***
  Reviewed 15 manuscripts (2024 – 2026)
- **Reviewer for *F1000Research*** *[Ad hoc]*
  Reviewed 1 manuscript (2024)

TECHNICAL STRENGTHS

- **Simulation & Modeling:** FlexSim [advanced], FlexSim HC, and AnyLogic softwares
- **Machine Learning & AI:** Machine learning, deep learning, reinforcement learning, data analytics
- **Programming Language:** Python, MATLAB, C/C++, R
- **Programming Libraries & Tools:** NumPy, Pandas, Scikit-learn, Keras, TensorFlow
- **Statistical Software:** SPSS, Minitab
- **3D Modeling and Design:** AutoCAD, SolidWorks

THEORITICAL STRENGTHS

Introduction to computational science; Numerical analysis; Mathematical and Computer Modelling; Advanced scientific computing; Advanced Algorithms; Computational Methods of Linear Algebra; Computer Simulation Application; Industrial Data Analytics; Stochastic Processes; Machine Learning; Deep Learning; Time Series Analysis; Statistical Analysis; Deterministic Decision Models; Stochastic Decision Models; Combinatorial Optimization; Computer-integrated Manufacturing Systems; Modern Manufacturing Processes, Operations Research, Operations Management

HONORS & AWARDS

- NSF Student Researcher Travel Award in 2026
- Conference Travel Fund from the College of Science in 2026
- **2nd Place (Runner-Up)**, FlexSim Student Simulation Competition, Healthcare Systems Process Improvement (HSPI) Conference, 2024
- Technical Education Scholarship in KUET (2013 - 2016)
- Scholarship awarded by a Bengali community organization in Germany (2010–2016)
- Board scholarship for academic excellence in 2010, Jessore board, Bangladesh.

PROFESSIONAL SERVICES

Logistics Secretary, **Bangladeshi Student Association (BSA), University of Texas at El Paso (UTEP)**
2025–2026

- Support organizational logistics and coordination for academic, cultural, and community events.

Student Mentor, MARCS Tutoring Center, University of Texas at El Paso (UTEP)
Fall 2024 & Spring 2025

- Provided academic support and problem-solving assistance to students in mathematics-related subjects.

REFERENCES

- **Dr. Ming-Ying Leung**
  Director
  Computational Science and Bioinformatics Programs
  University of Texas at El Paso
  Email: mleung@utep.edu
  Phone: (915) 747-6836
- **Dr. Md Fashiar Rahman**
  Assistant Professor
  Department of Industrial, Manufacturing, and Systems Engineering
  University of Texas at El Paso
  Email: mrahman13@utep.edu
  Phone: (915) 747-6903
- **Dr. Tzu-Liang (Bill) Tseng**
  Chair & Professor
  Department of Industrial, Manufacturing, and Systems Engineering
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